A Threefold-Symmetric Yu–Oh Context Diagram

(Dated: October 22, 2026)

Figure 1 retains the threefold Yu–Oh layout [1, 2] and the current vertex numbering. Smooth dashed curves continue the solid context lines at their endpoints, with matching tangents, to the common apex.

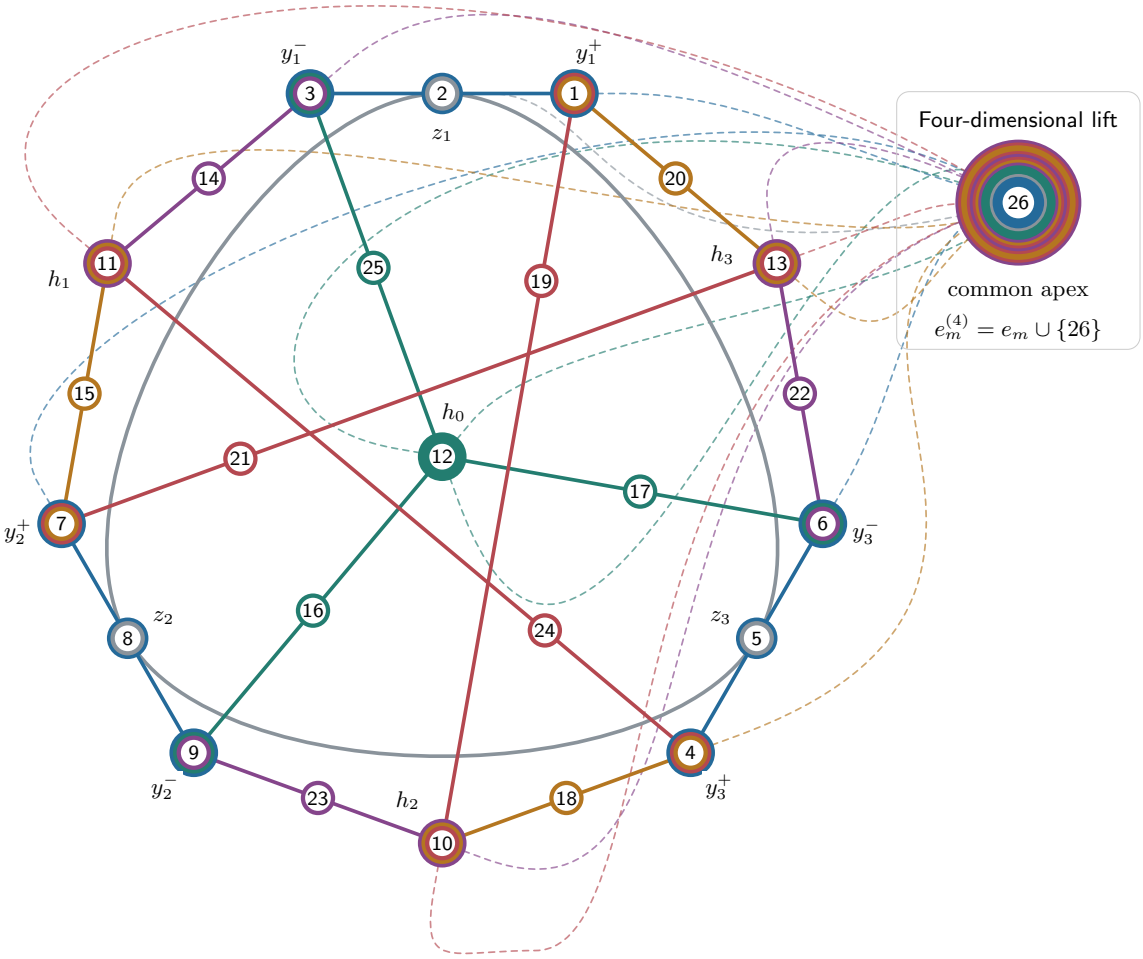


FIG. 1. Yu–Oh context diagram and its apex suspension. The base has 25 vertices and 16 three-element contexts. Each dashed curve leaves the end of its straight context trace with the same outward tangent and joins apex 26, displaying a four-element context. The grey coordinate context $\{2, 5, 8\}$ joins the apex tangentially at vertex 2. Unmarked crossings are not vertices. Colours identify the six context orbits under the 120° rotation; rings record incident context colours. Traditional symbols label the original 13 rays.

The 120° rotation fixes $12 = h_0$ and cycles the remaining base vertices in eight triples. On context indices it acts as (1 2 3)(5 6 7)(8 11 16)(9 13 14)(10 12 15), with context 4 fixed. The dashed continuations record the membership of the fixed apex 26 in every context; the base geometry and its colour symmetry are unchanged.

[1] K. Svozil, Chromatic quantum contextuality, Entropy **27**, 387 (2025), arXiv:2501.15261.

[2] S. Yu and C. H. Oh, State-independent proof of Kochen-Specker theorem with 13 rays, Physical Review Letters **108**, 030402 (2012), arXiv:1109.4396.